



Lists

Practical Skills Module 2: Topic 6

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Topic Code	PM-02-PS06:
Topic	Use Python lists to store multiple items in a single variable

Scope of Practical Skill

Given an applicable instruction and access to a learning platform, the learner must be able to:

- PA0601 Create lists
- PA0602 Access list elements
- PA0603 List slicing
- PA0604 Add/Change List Elements
- PA0605 Delete list elements
- PA0606 Find AVERAGE of a List in Python
- PA0607 Remove elements from Python List with [clear, POP, remove, del]
- PA0608 Concatenate variables

Applied Knowledge

- AK0601 Concept, definition and functions of Python data types

Internal Assessment Criteria

- IAC0601 Expected results with Python list are achieved

Resources:

Knowledge	Information from Python Programmer Curriculum
National Curriculum Framework	900221-000-00-PM-02



Practical task

Follow the facilitator's instructions to complete the following activities:

Scenario: Student Grades Management System

You are tasked with developing a Python program for a school that manages student grades. The program will allow the user to perform various operations on a list of grades, such as accessing specific grades, modifying them, removing grades, and calculating the average.

Create a Python file called **student_grades.py** and complete the following steps:

Step 1: Create a list of student grades.

- Create a list that contains at least **five numeric grades** as integers or floats.
- Store the list in a variable with a descriptive name (e.g., `grades_list`).

Tip: Use values that reflect actual grades (e.g., between 0 and 100).

Step 2: Access individual elements by index.

- Use indexing to **print the first and last grade** from the list.
- Include a label for each printed value so the output is clear.
 - *Example output:*

First grade in list: 88

Last grade in list: 76

Step 3: Use list slicing.

- Extract and print a **subset of the list** containing the grades of the **top three students**.
- Use slicing notation to get the first three elements.
 - *Expected output:* A sublist with the first three grades.

Step 4: Add and update elements.

- **Add a new grade** to the end of the list using the appropriate list method.
- **Update one of the existing grades** by assigning a new value to a specific index.

Tip: Choose any index and replace that grade with a new value.

Step 5: Delete a specific element.

- Remove one grade from the list by specifying its index using the `del` keyword.
- Print the updated list to confirm the change.

Step 6: Calculate the average of the grades.

- Use the `sum()` function and the `len()` function to calculate the **average** of the grades.
- Format the average to **two decimal places** and print it.
 - *Example output:* Average grade: 78.56

Step 7: Remove elements using different methods.

- Demonstrate at least **three of the following methods** to remove grades from the list:
 1. `pop()` — remove by index
 2. `remove()` — remove by value
 3. `clear()` — remove all items
 4. `del` — remove by index or slice
- Print the list after each removal so you can see the effect of each method.

Tip: You may want to re-initialise the list before each method for testing.

Step 8: Concatenate two lists.

- Create a second list with additional grades.
- Combine the original list with the new one into a single list using list concatenation.
- Print the final combined list.
 - *Example output:* Combined grades list: [85, 90, 78, 92, 88, 79, 84]



Take note

Important: Be sure to upload all files required for the task submission inside your task folder and then click "Request review" on your dashboard.
